Systems of Linear Equations (AA HL)

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two independent random variables are normally distributed as $X \sim N(50, 16)$ and $Y \sim N(45, 9)$. Find the probability that $X > Y$.

1. Define the difference variable $W = X - Y$. The condition $X > Y$ is equivalent to $W > 0$. 2. Since X and Y are independent normals, W is normally distributed: Mean: $\mu_W = E[X] - E[Y] = 50 - 45 = 5$. Variance: $\sigma_W^2 = \text{Var}(X) + \text{Var}(Y) = 16 + 9 = 25 \Rightarrow \sigma_W = 5$. Thus $W \sim N(5, 25)$. 3. Compute $P(W > 0)$: $z = \frac{0-5}{5} = -1.0$. $P(W > 0) = P(Z > -1.0) = P(Z < 1.0) = \Phi(1.0) \approx 0.8413 \approx 0.841$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the points $A(1, -2, 3)$, $B(4, 0, 5)$, and $C(p, 1, 2)$. If the vectors $\vec{AB}$ and $\vec{AC}$ are perpendicular, find the value of p.

First, find the vector $\vec{AB}$: $\vec{AB} = B - A = (4 - 1, 0 - (-2), 5 - 3) = (3, 2, 2)$. Next, find the vector $\vec{AC}$: $\vec{AC} = C - A = (p - 1, 1 - (-2), 2 - 3) = (p - 1, 3, -1)$. Two vectors are perpendicular if their dot product is zero. Therefore, $\vec{AB} \cdot \vec{AC} = 0$. $(3)(p - 1) + (2)(3) + (2)(-1) = 0$ $3p - 3 + 6 - 2 = 0$ $3p + 1 = 0$ $3p = -1$ $p = -\frac{1}{3}$ Final Answer: $p = 4$

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A fair six-sided die is rolled. What is the probability of rolling a number greater than 4?

The possible outcomes when rolling a fair six-sided die are {1, 2, 3, 4, 5, 6}. There are 6 equally likely outcomes in total. The outcomes that are greater than 4 are {5, 6}. There are 2 favorable outcomes. The probability of an event is the ratio of the number of favorable outcomes to the total number of possible outcomes. Therefore, the probability of rolling a number greater than 4 is $\frac{2}{6} = \frac{1}{3}$. Final Answer: 1/3

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Phone calls arrive at an emergency center according to a Poisson process with an average rate of 3 calls per minute. Find: (a) the probability that at least 2 calls arrive in a 1-minute period; (b) the probability that exactly 5 calls arrive in a 2-minute period.

(a) For 1 minute, $X \sim \text{Po}(3)$: $P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - e^{-3}\frac{3^0}{0!} - e^{-3}\frac{3^1}{1!} = 1 - e^{-3}(1 + 3) = 1 - 4e^{-3} \approx 1 - 0.19915 = 0.8009 \approx 0.801$. (b) For 2 minutes, the rate scales linearly: $\lambda = 3 \times 2 = 6$, so $Y \sim \text{Po}(6)$: $P(Y = 5) = \frac{e^{-6} \cdot 6^5}{5!} = \frac{e^{-6} \cdot 7776}{120} = 64.8e^{-6} \approx 0.1606 \approx 0.161$.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A discrete random variable X takes values $0, 3, 4$ with probabilities $\frac{1}{10}, \frac{3}{10}, \frac{3}{5}$ respectively. Find $E(X)$.

Working through this statistics step: $E(X) = \sum xP(X = x) = 0(\frac{1}{10}) + 3(\frac{3}{10}) + 4(\frac{3}{5}) = \frac{33}{10}$ Final Answer: $\frac{33}{10}$ Key Mathematical Principle & Verification: In exam revision, always verify that the final value satisfies the domain constraints, units, and original governing equation by direct back-substitution into the prompt.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The region bounded by $y = \ln x$, the line $y = 0$, the line $y = 1$, and the y-axis is rotated 2π radians about the y-axis. Find the exact volume of the resulting solid.

1. Express x as a function of y: $y = \ln x \Rightarrow x = e^y$. 2. Apply the volume of revolution about the y-axis formula: $V = \pi\int_c^d [x(y)]^2\,dy = \pi\int_0^1 (e^y)^2\,dy = \pi\int_0^1 e^{2y}\,dy$. 3. Integrate: $V = \pi\left[\frac{e^{2y}}{2}\right]_0^1 = \frac{\pi}{2}(e^2 - e^0) = \frac{\pi(e^2-1)}{2}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = \frac{1}{2}x^2 - 3x + 5$. The graph of $y = f(x)$ is a parabola. Find the x-coordinate of the vertex of the parabola.

The x-coordinate of the vertex of a parabola given by the equation $f(x) = ax^2 + bx + c$ is found using the formula $x = -\frac{b}{2a}$. In this case, $a = \frac{1}{2}$ and $b = -3$. Therefore, the x-coordinate of the vertex is $x = -\frac{-3}{2(\frac{1}{2})} = -\frac{-3}{1} = 3$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find $\int xe^{2x}\,dx$ (in terms of x, with constant C).

Working through this calculus step: $u = x,\ dv = e^{2x}dx \Rightarrow \int xe^{2x}dx = \frac{x}{2}e^{2x} - \int \frac{1}{2}e^{2x}dx = \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$ Final Answer: $\frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the number of solutions of $\sin(2\theta) = \frac{1}{2}$ in the interval $0^\circ \le \theta < 360^\circ$.

Final Answer: 4. Working through this additional-topics step:

$2\theta = 30^\circ, 150^\circ, 390^\circ, 510^\circ$ (within $0^\circ \le 2\theta < 720^\circ$) $\Rightarrow \theta = 15^\circ, 75^\circ, 195^\circ, 255^\circ$: 4 solutions.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

How many solutions does $\sin\theta = \sin 30^\circ$ have in the interval $0^\circ \le \theta < 360^\circ$? Show your complete algebraic work.

Final Answer: 2. Working through this additional-topics step:

$\theta = 30^\circ$ or $\theta = 180^\circ - 30^\circ = 150^\circ$: 2 solutions in $[0^\circ, 360^\circ)$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

A network has weighted adjacency matrix W. Explain how the minimum spanning tree algorithm (Kruskal's) differs from Dijkstra's algorithm.

Kruskal's algorithm finds a global minimum spanning tree that connects all vertices in the graph with minimum total edge weight, without reference to any single starting vertex. Dijkstra's algorithm finds the shortest path routes from a designated single source vertex to all other individual vertices. A minimum spanning tree does not necessarily contain the shortest paths between arbitrary pairs of vertices.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

A student receives the following scaled marks on IB AA HL assessment: Paper 1: 85/110, Paper 2: 92/110, Paper 3: 44/55, and IA: 16/20. Calculate the student's overall weighted percentage.

1. Compute the component percentages: Paper 1: $\frac{85}{110} \approx 77.27\%$ Paper 2: $\frac{92}{110} \approx 83.64\%$ Paper 3: $\frac{44}{55} = 80.00\%$ IA: $\frac{16}{20} = 80.00\%$ 2. Multiply by component weights (30%, 30%, 20%, 20%):

$\text{Overall} = 0.30(77.27) + 0.30(83.64) + 0.20(80.00) + 0.20(80.00)$ $\text{Overall} = 23.181 + 25.091 + 16.000 + 16.000 = 80.27\% \approx 80.3\%.$

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A continuous random variable X has probability density function $f(x) = 2e^{-2x}$ for $x \geq 0$ and 0 elsewhere. Find the cumulative distribution function $F(x)$ and determine the exact median of X.

1. Find CDF for $x \geq 0$: $F(x) = \int_0^x 2e^{-2t}\,dt = [-e^{-2t}]_0^x = 1 - e^{-2x}.$ 2. Set $F(m) = 0.5$ to find the median m: $1 - e^{-2m} = 0.5 \Rightarrow e^{-2m} = 0.5 \Rightarrow -2m = \ln(0.5) = -\ln 2 \Rightarrow m = \frac{\ln 2}{2}.$

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Find the perpendicular distance from point $P(3, 1, -2)$ to plane $\pi : 2x - 2y + z = 5$.

Apply the point-to-plane distance formula: $D = \frac{|aP_x + bP_y + cP_z - d|}{\sqrt{a^2 + b^2 + c^2}} = \frac{|2(3) - 2(1) + 1(-2) - 5|}{\sqrt{2^2 + (-2)^2 + 1^2}} = \frac{|6 - 2 - 2 - 5|}{\sqrt{4 + 4 + 1}} = \frac{|-3|}{3} = 1.$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra**

Find the volume of the solid generated when the region bounded by $y = \sqrt{x}$, the x-axis, and $x = 4$ is revolved 2π radians about the x-axis.

Apply the volume of revolution formula: $V = \pi\int_0^4 y^2\,dx = \pi\int_0^4 (\sqrt{x})^2\,dx = \pi\int_0^4 x\,dx = \pi\left[\frac{x^2}{2}\right]_0^4 = \pi\left(\frac{16}{2} - 0\right) = 8\pi.$

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16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find the Maclaurin series for $\ln(1+2x)$ up to the x^3 term, and state its interval of convergence.

Use $\ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \ldots$ with $u = 2x$: $\ln(1+2x) = 2x - 2x^2 + \frac{8}{3}x^3 - \ldots$. Convergence requires $-1 < 2x \leq 1 \Rightarrow -\frac{1}{2} < x \leq \frac{1}{2}$.

17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.04 Partial Fractions

[AA HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

[AI HL — Additional Topics · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

[Overview · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

[Functions (HL Extensions) · Practice 2] If the governing relation for this system is $g(t) = 4t + 4$, evaluate $g(6)$ and find the inverse input t when $g(t) = 32$.

Substitute $t = 6$: $g(6) = 4(6) + 4 = \mathbf{28}$. Setting $4t + 4 = 32$ gives $4t = 28 \Rightarrow t = \mathbf{7}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In row-reducing an augmented matrix, which row operation is NOT permitted?

a. Squaring all elements in a row ✓

b. Swapping two rows

c. Multiplying a row by a non-zero scalar

d. Adding a multiple of one row to another row

The three valid elementary row operations are row swaps, multiplying a row by a non-zero constant, and adding a scalar multiple of a row to another row. Squaring entries alters the solution space.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 5]$. What does this indicate about the system?

a. The system is inconsistent (has no solutions) ✓

b. The system has infinitely many solutions

c. The system has a unique solution with $z = 5$

d. The system has trivial solutions only

The row represents the contradictory equation $0x + 0y + 0z = 5$ (i.e. $0 = 5$), which cannot be satisfied by any real values; hence the system is inconsistent.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

An augmented matrix in row echelon form has final row $[0, 0, 0 \mid 0]$ with 2 leading pivots across 3 variables. What does this signify?

a. The system has infinitely many solutions parameterized by 1 free variable ✓

b. The system has a unique solution

c. The system has no solutions

d. The determinant of the matrix is 1

With 3 variables and 2 leading pivot columns, there is $3 - 2 = 1$ free variable, which gives a line of infinitely many solutions.

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24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the geometric configuration of three linear equations in three variables that possess a unique solution?

a. Three planes intersecting at a single unique point ✓

b. Three planes intersecting along a common line

c. Three parallel non-intersecting planes

d. Three planes forming a triangular prism

A unique solution (x_0, y_0, z_0) means all three planes share exactly one point of intersection.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If three planes intersect along a single straight line, what type of system is this?

a. Consistent dependent system with infinitely many solutions ✓

b. Inconsistent system with no solutions

c. Consistent independent system with 1 solution

d. Homogeneous system with trivial solution only

Intersection along a line means infinitely many common points exist, parameterized by 1 free variable.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Three planes form the faces of an open triangular prism with no common line or point. How is this system classified?

a. Inconsistent (no solutions) ✓

b. Infinitely many solutions

c. Unique solution

d. Singular dependent

Each pair of planes intersects in a line, but all three lines are parallel and distinct, meaning there is no point common to all three planes simultaneously.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate the 2×2 determinant $\det[[4, -3], [2, 5]]$.

a. 26 ✓

b. 14

c. -26

d. 20

$\det = ad - bc = 4(5) - (-3)(2) = 20 - (-6) = 26$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Under what condition does a 3×3 system $A\mathbf{x} = \mathbf{b}$ have a unique solution?

a. $\det(A) \neq 0$ ✓

b. $\det(A) = 0$

c. $\mathbf{b} = \mathbf{0}$

d. $\text{Tr}(A) > 0$

A square system $A\mathbf{x} = \mathbf{b}$ has a unique solution if and only if the coefficient matrix A is non-singular, meaning $\det(A) \neq 0$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\det(A) = 0$ for a system $A\mathbf{x} = \mathbf{b}$, what can be concluded?

a. **The system either has no solutions or infinitely many solutions** ✓

b. The system must have no solutions

c. The system must have infinitely many solutions

d. The system has exactly 2 solutions

When $\det(A) = 0$, A is singular; depending on the consistency of the constant vector $\mathbf{b}$, the system has either zero solutions or infinitely many solutions.

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30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Solve the system: $x + y + z = 6$ and $2x - y + z = 3$. Parameterize the solution line using $z = \lambda$.

a. $x = 3 - \frac{2}{3}\lambda, y = 3 - \frac{1}{3}\lambda, z = \lambda$ ✓

b. $x = 3 + \lambda, y = 3 - \lambda, z = \lambda$

c. $x = 9 - 2\lambda, y = \lambda, z = \lambda$

d. $x = 1, y = 2, z = 3$

Add equations: $3x + 2z = 9 \Rightarrow 3x = 9 - 2\lambda \Rightarrow x = 3 - \frac{2}{3}\lambda$. Subtract from first: $y = 6 - x - z = 6 - (3 - \frac{2}{3}\lambda) - \lambda = 3 - \frac{1}{3}\lambda$.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For what value of k does the system $x + 2y = 4$ and $3x + 6y = k$ have infinitely many solutions?

a. $k = 12$ ✓

b. $k = 4$

c. $k = 0$

d. $k = 6$

Multiply the first equation by 3: $3x + 6y = 12$. For infinitely many solutions, the equations must be identical, so $k = 12$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If the homogeneous system $A\mathbf{x} = \mathbf{0}$ has non-trivial solutions (solutions other than $\mathbf{x} = \mathbf{0}$), what must be true about $\det(A)$?

a. $\det(A) = 0$ ✓

b. $\det(A) > 0$

c. $\det(A) = 1$

d. $\det(A) < 0$

A homogeneous linear system has non-trivial solutions if and only if the coefficient matrix is singular, i.e., $\det(A) = 0$.

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33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagram shows the graph of the function $f(x) = 2x^2 - 8x + k$. The graph intersects the x-axis at points A and B, and the vertex of the parabola is at point V. Given that the x-coordinate of the vertex is 2, and the y-coordinate of the vertex is -6, find the value of k.

a. 1

b. -2

c. **2 ✓**

d. -1

The vertex of a parabola given by $f(x) = ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $f(x) = 2x^2 - 8x + k$, so $a = 2$ and $b = -8$. The x-coordinate of the vertex is given as 2, which is consistent: $x = -\frac{-8}{2(2)} = \frac{8}{4} = 2$. The y-coordinate of the vertex is $f(2)$. We are given that the y-coordinate of the vertex is -6. Therefore, $f(2) = -6$. Substitute $x = 2$ into the function: $f(2) = 2(2)^2 - 8(2) + k$. $f(2) = 2(4) - 16 + k$ $f(2) = 8 - 16 + k$ $f(2) = -8 + k$. Since $f(2) = -6$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is a parabola. Find the coordinates of the vertex of the parabola.

a. **(2, -1) ✓**

b. (-2, 1)

c. (2, 1)

d. (-2, -1)

The function is given by $f(x) = x^2 - 4x + 3$. This is a quadratic function in the form $ax^2 + bx + c$, where $a = 1$, $b = -4$, and $c = 3$. The x-coordinate of the vertex of a parabola is given by the formula $x = -\frac{b}{2a}$. Substituting the values of a and b, we get $x = -\frac{-4}{2(1)} = \frac{4}{2} = 2$. To find the y-coordinate of the vertex, substitute this x-value back into the function: $f(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the coordinates of the vertex are (2, -1).

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35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the function $f(x) = x^2 - 6x + 8$. The graph of $y = f(x)$ intersects the x-axis at points A and B, and the y-axis at point C. Find the length of the line segment AB.

a. 2 units ✓

b. 3 units

c. 4 units

d. 6 units

To find the points where the graph intersects the x-axis, we set $f(x) = 0$. $x^2 - 6x + 8 = 0$. Factoring the quadratic equation, we get $(x - 2)(x - 4) = 0$. Thus, the x-intercepts are $x = 2$ and $x = 4$. Let these points be A and B. The distance between A and B is $|4 - 2| = 2$ units. To find the y-intercept, we set $x = 0$. $f(0) = 0^2 - 6(0) + 8 = 8$. So, point C is (0, 8). The question asks for the length of the line segment AB, which is the distance between the x-intercepts.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The diagram shows the graph of the function $f(x) = x^2 - 6x + k$. The vertex of the parabola is at the point $(h, -1)$. Determine the value of k.

a. 8 ✓

b. 7

c. 6

d. 5

The given function is $f(x) = x^2 - 6x + k$. The x-coordinate of the vertex of a parabola in the form $ax^2 + bx + c$ is given by $h = -\frac{b}{2a}$. In this case, $a = 1$ and $b = -6$, so $h = -\frac{-6}{2(1)} = 3$. The vertex is given as $(h, -1)$, so we know $h = 3$. To find k, we substitute the coordinates of the vertex into the function: $f(3) = -1$. $(3)^2 - 6(3) + k = -1$ $9 - 18 + k = -1$ $-9 + k = -1$ $k = -1 + 9$ $k = 8$. Therefore, the value of k is 8.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is a parabola. Find the x-coordinates of the points where the graph intersects the x-axis.

a. 1 and 2 ✓

b. -1 and -2

c. 2 and 3

d. -1 and 2

To find the x-coordinates of the points where the graph intersects the x-axis, we need to find the values of x for which $f(x) = 0$. Set the function equal to zero: $x^2 - 3x + 2 = 0$ Factor the quadratic equation: $(x - 1)(x - 2) = 0$ This gives two possible solutions for x: $x - 1 = 0 \Rightarrow x = 1$ $x - 2 = 0 \Rightarrow x = 2$ Therefore, the graph intersects the x-axis at $x = 1$ and $x = 2$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B. Determine the distance between A and B.

a. **2** ✓

b. 3

c. 4

d. 5

To find the x-intercepts, we set $f(x) = 0$. Thus, we solve the quadratic equation $x^2 - 4x + 3 = 0$. This can be factored as $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. These are the x-coordinates of points A and B. Therefore, the points are (1, 0) and (3, 0). The distance between these two points on the x-axis is the absolute difference of their x-coordinates: $|3 - 1| = 2$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. Determine the value of $f'(2)$.

a. -2

b. **0** ✓

c. 2

d. 4

The function is given by $f(x) = -x^2 + 4x - 3$. To find the derivative $f'(x)$, we differentiate each term with respect to x: $f'(x) = \frac{d}{dx}(-x^2) + \frac{d}{dx}(4x) - \frac{d}{dx}(3)$ $f'(x) = -2x + 4 - 0$ $f'(x) = -2x + 4$. To find the value of $f'(2)$, we substitute $x = 2$ into the derivative: $f'(2) = -2(2) + 4$ $f'(2) = -4 + 4$ $f'(2) = 0$. Therefore, the value of $f'(2)$ is 0.

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40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = x^2 - 4x + 3$ is shown below. The graph intersects the x-axis at points A and B. The vertex of the parabola is at point C. Determine the coordinates of the point where the graph intersects the y-axis.

a. **(0, 3)** ✓

b. (3, 0)

c. (0, -1)

d. (-1, 0)

To find the y-intercept of the graph of a function, we set $x = 0$ in the function's equation. For the function $f(x) = x^2 - 4x + 3$, substituting $x = 0$ gives: $f(0) = (0)^2 - 4(0) + 3$ $f(0) = 0 - 0 + 3$ $f(0) = 3$ Therefore, the graph intersects the y-axis at the point (0, 3).

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

The diagram shows the graph of the function $f(x) = 3x^2 + bx - 6$. The graph passes through the point $(1, 1)$. Find the value of b.

a. 3

b. **4** ✓

c. 5

d. 6

The function is given by $f(x) = 3x^2 + bx - 6$. Since the graph passes through the point $(1, 1)$, we can substitute $x = 1$ and $f(x) = 1$ into the equation: $1 = 3(1)^2 + b(1) - 6$ $1 = 3 + b - 6$ $1 = b - 3$ Adding 3 to both sides, we get: $b = 1 + 3$ $b = 4$

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

Consider the function $f(x) = -x^2 + 4x - 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at two points. What is the value of the x-coordinate of the point of intersection with the larger x-value?

a. 1

b. 2

c. **3** ✓

d. 4

To find the x-intercepts, we set $f(x) = 0$. This gives the quadratic equation $-x^2 + 4x - 3 = 0$. Multiplying by -1, we get $x^2 - 4x + 3 = 0$. Factoring this equation, we have $(x - 1)(x - 3) = 0$. The solutions are $x = 1$ and $x = 3$. The larger x-value is 3.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.02 The Poisson Distribution

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The graph intersects the x-axis at points A and B, and the y-axis at point C. Find the coordinates of the point where the graph of $y = f(x)$ intersects the y-axis.

a. **(0, 3)** ✓

b. (3, 0)

c. (1, 0)

d. (0, -3)

To find the y-intercept, we need to find the value of $f(x)$ when $x = 0$. \\ Substituting $x = 0$ into the function: \\ \(f(0) = (0)^2 - 4(0) + 3 \\ f(0) = 0 - 0 + 3 \\ f(0) = 3 \\ Therefore, the graph intersects the y-axis at the point (0, 3).

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Further Integration Techniques

Consider the function $f(x) = x^2 - 3x + 2$. The graph of $y = f(x)$ is shown. Determine the y-coordinate of the y-intercept.

a. 1

b. **2** ✓

c. 0

d. -1

The y-intercept occurs when $x = 0$. Substituting $x = 0$ into the function: $f(0) = (0)^2 - 3(0) + 2 = 0 - 0 + 2 = 2$. Therefore, the y-coordinate of the y-intercept is 2.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.02 Differential Equations

A particle's position in the complex plane is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$, where t is time in seconds and $t \geq 0$. The particle starts at $t = 0$ and its path is traced by the complex number $z(t)$. The particle is said to be 'at rest' when its velocity, represented by the complex number $v(t) = z'(t)$, is zero. Determine the times $t > 0$ when the particle is at rest.

a. $t = 1$ and $t = 3$

b. $t = 1$ and $t = 2$

c. $t = 2$ and $t = 3$

d. $t = 1, t = 2,$ **and** $t = 3$ ✓

The particle's position is given by $z(t) = (2t^2 - 3t + 1) + i(t^3 - 6t^2 + 11t - 6)$. The velocity is the derivative of the position with respect to time, $v(t) = z'(t)$. Differentiating the real and imaginary parts separately: Real part of $z'(t)$: $\frac{d}{dt}(2t^2 - 3t + 1) = 4t - 3$ Imaginary part of $z'(t)$: $\frac{d}{dt}(t^3 - 6t^2 + 11t - 6) = 3t^2 - 12t + 11$ So, $v(t) = (4t - 3) + i(3t^2 - 12t + 11)$. The particle is at rest when $v(t) = 0$. This means both the real and imaginary parts of $v(t)$ must be.

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